

Raied Ahmed Nishat

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EDUCATION

M.S. in Civil Engineering

Georgia Southern University, Statesboro, GA, USA

Advisor: [Dr. Ram Krishna Mazumder](#)

August 2026 - Present

B.Sc (Engg.) in Civil and Environmental Engineering

Shahjalal University of Science and Technology (SUST), Sylhet 3114, Bangladesh

GPA: 3.64/4.00, Class Rank 5/55

Undergraduate Thesis: A Transfer Learning Approach for Soil Strength Prediction

Advisor: [Dr. Shriful Islam](#)

Jan 2020 - Sep 2025

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Department of Civil Engineering and Construction

Georgia Southern University, Statesboro, GA

- Conduct research on water distribution system asset management, focusing on predictive modeling and risk assessment of critical water infrastructure.
- Support capital improvement planning and research activities aimed at improving the reliability, resilience, and long-term sustainability of water distribution systems.

August 2026 - Present

Lecturer (Adjunct)

Department of Civil Engineering, Leading University, Sylhet-3112, Bangladesh

- Taught Structural Mechanics, covering stress analysis, beam deflection, and column buckling.
- Conducted Civil Engineering laboratory sessions in Open Channel Flow Sessional and Environmental Engineering Sessional II, focusing on hydraulic experiments, water/wastewater systems, and EPANET modeling.
- Taught and supervised design courses including Concrete Structures Sessional II, Traffic Planning and Management, and Building Services II: Plumbing, covering structural design, transportation planning, and building water/sewer systems.

Jan 2026 – August 2026

Junior Project Engineer

Center for Research Testing and Consultancy (CRTC), SUST, Sylhet-3114

- Supervised on-site SPT operations at Sylhet Gas Field sites and conducted laboratory soil classification and strength testing to ensure reliable geotechnical data.
- Contributed to technical analysis and preparation of engineering reports for consultancy projects.

Dec 2025 - Jan 2026

HONORS AND ACHIEVEMENTS

- Semifinalist, Hult Prize at SUST, 2023
- Chandpur District Parishad Higher Secondary Merit Scholarship
- Merit Scholarship, Secondary School Certificate
- Chandpur District Parishad Secondary School Merit Scholarship
- Ashutosh Education Scholarship, Brahmanbaria

RESEARCH EXPERIENCE

Research Assistant

Department of Civil and Environmental Engineering, SUST

Advisor: Prof. Dr. Shriful Islam

Apr 2024 - July 2026

- Laboratory-based experimental and computational studies on material and soil characterization.
- Applying machine learning in geotechnical modeling of soil strength and relevant parameters.
- Conducting characterization tests, data analysis, report and manuscript preparation for ongoing projects.

Collaborative Research Projects (journal manuscript in-preparation)

1. Performance Evaluation of Water Distribution Systems

Co-Authors: [Dr. Ram Krishna Mazumder](#) (Georgia Southern University)

Topic: This study evaluates various performance metrics, including topographic, hydraulic, water security and economic metrics for water networks.

2. Community-Level Scenario Seismic Risk and Resilience Analysis of Chittagong

Co- Authors: [Abul Hasnat](#) (Texas A&M University) and [Dr. Ram Krishna Mazumder](#) (Georgia Southern University)

Topic: Using scenario seismic hazards, this study evaluates seismic risk of buildings and post-earthquake recovery for Chittagong city. Utilizing the HAZUS framework, this study incorporates uncertainty considerations in damage, retrofitting, and recovery simulations.

3. **AI-Enabled Sewer Defect Detection Using CCTV Inspection Data for Asset Management Planning**
Co-Authors: [Julker Nayen Rafi](#) and [Dr. Ram Krishna Mazumder](#) (Georgia Southern University)
Topics: Developing an Artificial Intelligence assisted asset management framework that utilizes automated sewer inspection data for actionable engineering decision-making to prioritize rehabilitation efforts.
4. **Machine Learning in Soil Strength Prediction: A Systematic Review**
Co-Authors: Fahmida Rahman (SUST), and Dr. Shfirul Islam (SUST)
Topic: First targeted systematic review of machine learning application in soil strength parameter prediction. Following PRISMA guided methodology for systematic reviews, this study compiles a comprehensive review of current literature trends, gaps, providing future directions for this field.
5. **Uncertainty Aware Bayesian Physics Informed Neural Network for Probabilistic Slope Stability Classification**
Co-Authors: Nabid Hasnat (Leading University, Sylhet)
Topics: Incorporating monotonic physics constraints in Bayesian Neural Networks for developing a physically consistent, uncertainty-aware machine learning framework for slope stability classification.

PUBLICATIONS

Journal Articles

1. **Nishat, R. A.**, Rahman, F., Hoque, N. M. R., & Islam, S. (2026). Transfer learning for undrained shear strength prediction: A case study with instance-based transfer. *Geosystems and Geoenvironment (Elsevier)*, 100519. <https://doi.org/10.1016/j.geogeo.2026.100519>
2. Islam S., **Nishat R. A.**, Sutradhar A., Hoque N. M. R., Rafi M. N. I. and Rana S. (202X). Effect of lime and alkali-activated LRF slag on geotechnical properties and microstructural behavior of kaolin clay, *International Journal of Geosynthetics and Ground Engineering (Springer)*. **(Accepted)**
3. Islam S., Chowdhury A.S., **Nishat R. A.**, Naba M.F., and Rafi M. N. I. (202X). Strength and mechanical behavior of fly ash and lime stabilized embankment soil reinforced with polyester fiber, *Indian Geotechnical Journal (Springer)*. **(Under Review)**
4. Joy M.R. and **Nishat R. A.** Hydroclimatic Trend Analysis and Machine Learning-Based Water Level Prediction for the Ganges (Padma) River Basin. *Acta Geophysica (Springer)*. **(Under Review)**
5. **Nishat R. A.** and Joy M.R. (2025). Investigating the Trends of Rainfall and Discharge along the Brahmaputra-Jamuna River System. *Technical Journal of River Research Institute*, 17(1). [\[ResearchGate\]](#)

Conference Papers

1. **Nishat R. A.**, Tusar A. N., Islam S., and Monni M. A. (2026). A Probabilistic Transformer-Based Machine Learning Model for Shallow Foundation Settlement Prediction. *8th International Conference on Civil Engineering for Sustainable Development, KUET, Khulna, 2026*. [\[ResearchGate\]](#)
2. **Nishat R. A.**, Rahman F., Hoque N. M. R., Rafi M. N. I and Islam S. (2026). Prediction Of Soil Unconfined Compressive Strength Using A Decision Tree-Based Hybrid Machine Learning Model. *8th International Conference on Advances in Civil Engineering 2026, CUET, Chattogram*

TECHNICAL SKILLS

- **Machine Learning Methods:** Neural Networks, Transfer learning, Gradient Boosting, Random Forest, Bayesian/Probabilistic Modeling
- **Geotechnical & Scientific Software:** EPANET, SWMM, AutoCAD, ETABS, ArcGIS, QGIS
- **Programming & Productivity:** Python, TensorFlow/Keras, Jupyter Notebook, LaTeX, C (basic), MATLAB (basic)

LEADERSHIP AND SERVICE ACTIVITY

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| General Secretary Aaj Muktomancho, SUST | 2024 – 2025 |
| <ul style="list-style-type: none"> • Led the 15th Executive Committee and served as convener of a 1,000+ attendee cultural concert, overseeing programs, partnerships, and public engagement. | |
| Organizing Secretary Association of Civil and Environmental Engineering, SUST | 2024 – 2025 |
| <ul style="list-style-type: none"> • Coordinated large-scale departmental events and facilitated collaboration among faculty, students, and industry stakeholders. | |

REFERENCES

Dr. Ram Krishna Mazumder

Assistant Professor, Dept. of Civil Engineering & Construction
Georgia Southern University
Statesboro, GA 30460, USA
rmazumder@georgiasouthern.edu
+1 912-478-0896
Relationship: Advisor, MS, Civil Engineering

Dr. Shriful Islam

Professor, Dept. of Civil & Environmental Engineering
Shahjalal University of Science and Technology
Sylhet-3114, Bangladesh
sharif-cee@sust.edu
+880 1716 687869
Relationship: Advisor, Undergraduate Research Project